

Romulus Khalil

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EDUCATION

Georgia Institute of Technology - Atlanta, GA

B.S. in Computer Science - Systems/Architecture and Information Internetworks

Graduation: Fall 2027

Major GPA: 3.6/4.0

SKILLS

Languages: C, Rust, Java, Lua, Python, TypeScript, PHP

Systems/Embedded: Linux kernel drivers, FPGA prototyping (Xilinx PYNQ-Z1), x86

AI/Data: LangGraph, A2A, AWS Bedrock, SQL

Tools/DevOps: Kubernetes, Helm, Flux, Docker, Nginx, Linux (Arch, Debian, RHEL), AWS (EC2)

EXPERIENCE

Technology Summer Analyst

Jun. 2026 – Aug. 2026

Morgan Stanley

Atlanta, GA

- Automated migration of 50+ legacy lending service configurations to the new credit risk platform structure with Shell and Python scripts, saving the team an estimated 2 weeks
- Built an AI personalization system with TypeScript, LangGraph, and Morgan Stanley's AI Gateway, persisting user context in SQL to dynamically augment agent prompts; adopted by the Middle Office team
- Migrated 10+ Lending workflows to LangGraph and A2A, building reusable TypeScript components, now used across 3 teams, to render agent-generated artifacts

Undergraduate Teaching Assistant

Aug. 2026 – Present

Design-Operating Systems

Atlanta, GA

- Guided 100+ students through core OS concepts: x86 boot procedures, virtual memory management (copy-on-write, zero pages), CPU scheduling, multithreading, filesystems, and networking
- Hold weekly office hours; wrote and graded 2 exams and 5 student projects

Undergraduate Researcher

Aug. 2025 – Present

Georgia Tech VIP: Configurable Computing/Embedded Systems Team

Atlanta, GA

- Developed a Linux driver for a custom 8-pin GPIO module (Verilog, built by the hardware team) on a dual-CPU secure voting machine prototype
- Hardened the low-side CPU driver so malicious button input cannot reach the security-critical high-side CPU, enforcing memory isolation between the two to protect information integrity
- Prototyped and tested on a Xilinx PYNQ-Z1 FPGA, building a test suite of 100+ cases covering input and memory faults

Software Engineering Intern

May 2025 – Aug. 2025

Azalea Health

Atlanta, GA

- Built an AI Chart Assistant for Azalea Health's hospital EHR product, using AWS Bedrock for secure access to foundation models including Claude Sonnet and Opus 4.5
- Wrote 12 production-tested PHP APIs used for context/initialization of the Chart Assistant, reducing prompt-to-response time by 50%
- Designed an agent-based session architecture with one agent per patient chart, isolating context to prevent context-window overflow

PROJECTS

The Home Server Project | cribothy.us | *Kubernetes, Flux, Debian*

June 2025 – Present

- Run a Kubernetes cluster hosting 20+ containerized services across media, monitoring, networking, and self-hosted applications
- Implemented GitOps-based infrastructure management with Flux and Kustomize, enabling declarative deployment and automated synchronization from a Git repository
- Configured dynamic NFS storage with a provisioner, then diagnosed recurring hangs as an NFS loopback deadlock (server and client pods on the same host) and moved stateful workloads to local-path PVCs
- Integrated NVIDIA GPU acceleration into Kubernetes using the NVIDIA Container Toolkit and device plugin, enabling GPU-aware scheduling for media transcoding

COURSEWORK & AWARDS

- Courses:** Advanced Operating Systems, Design & Analysis of Algorithms, Design-Operating Systems, Systems and Networking, Computer Organization & Programming, Data Structures & Algorithms
- Awards:** Zell Miller Scholarship Recipient, Dean's List 6x